

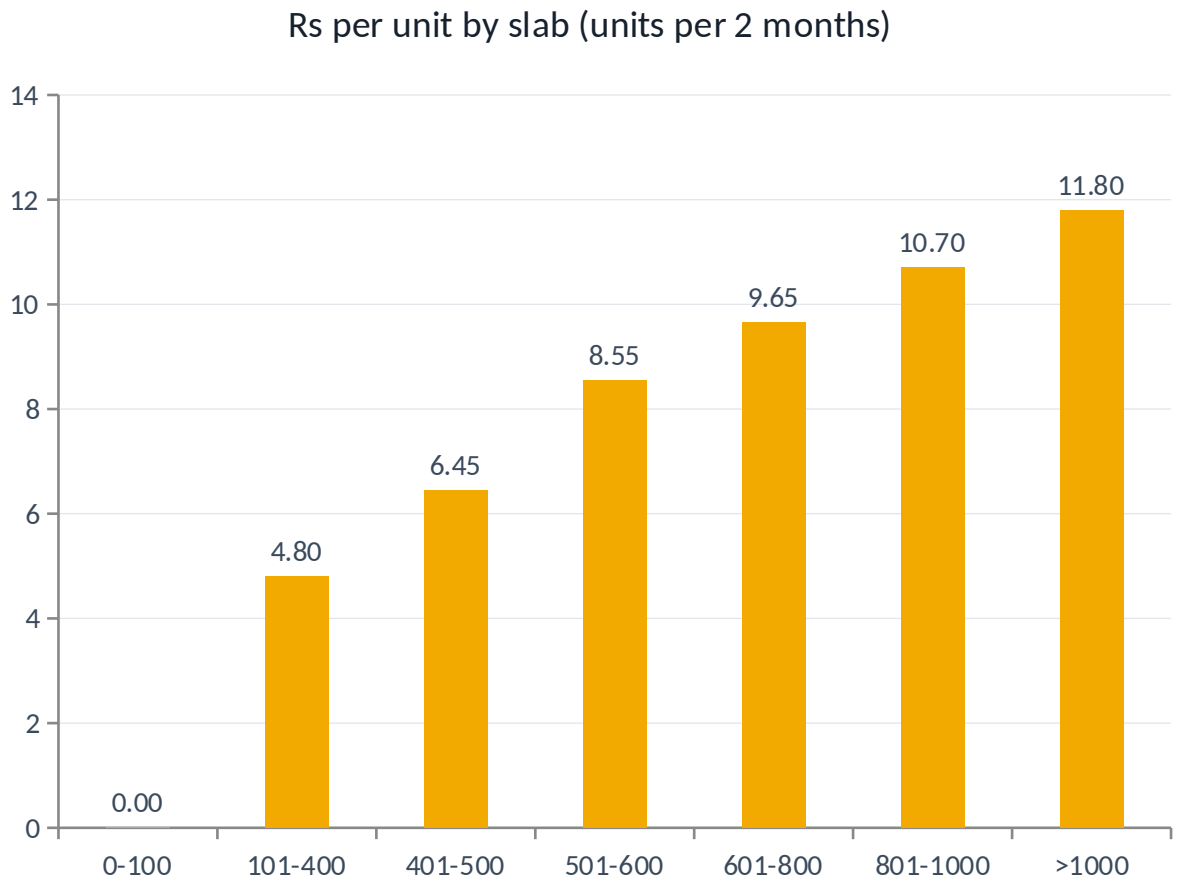
WATT-WATCH

Smart electricity bill warning and saving assistant for households

Presented by: [Your name] · [College / Department]

The problem: bills jump between slabs

- Tamil Nadu bills households every 2 months on a slab system
- The price of each extra unit rises as usage crosses each slab
- Families only see their usage when the bill arrives, too late to change it



Rates used in the app, based on TNERC Tariff Order No. 6 of 2025 (domestic LT IA). First 100 units free under the Government subsidy.

Our solution: see the bill coming



Scan
Photo of the meter or type the reading

Track
Units used since the last bill

Predict
Expected units and bill for the cycle

Warn
Early alert before a costlier slab

Reduce
Plan: which appliances to cut

Save
Units, money and CO2 saved

No extra hardware: a web app that works in any browser, on a phone or a computer.

Technology

Python

All calculations: bill, prediction, warning, saving plan

Streamlit

The web interface: welcome pop-up, sidebar and 4 tabs

SQLite

Database of households, meter readings and bills

Tesseract OCR

Reads the number from a meter photo

Database: three linked tables

Household
House_ID (PK)
Consumer_Number
Household_Name
Address
Connection_Type

→

EB_Meter
Meter_ID (PK)
House_ID (FK)
Meter_Number
Previous_Reading
Current_Reading
Units_Consumed
Reading_Date
Previous_Reading_Date *
Bill_ID (FK) *

→

Electricity_Bill
Bill_ID (PK)
House_ID (FK)
Billing_Period
Units_Consumed
Bill_Amount
Payment_Status

One household has many readings and many bills. Units_Consumed = Current_Reading − Previous_Reading (calculated, never typed). * Added: Previous_Reading_Date (to get units per day) and Bill_ID (empty = current, unbilled cycle).

Bill calculation: slab by slab

Slab (units)	Units	Rate (Rs)	Amount (Rs)
0 - 100	100	0.00	0.00
101 - 400	300	4.80	1,440.00
401 - 450	50	6.45	322.50
Total	**450**		**1,762.50**

Each unit is charged at the rate of the slab it falls in. Rates can be edited in the app's sidebar when the tariff changes.

Rs 1,762.50

Bill for 450 units in 2 months

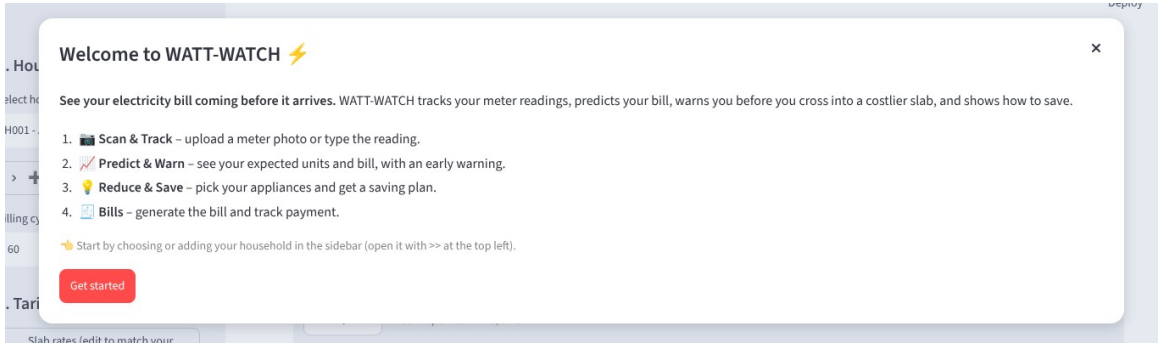
Rs 240

Sample: Arun, 150 units

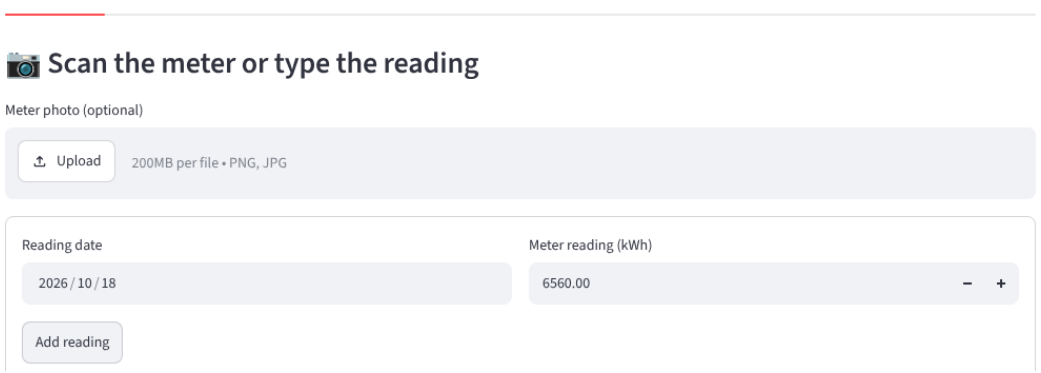
Rs 960

Sample: Lakshmi, 300 units

Welcome pop-up and meter tracking



A welcome pop-up explains the 4 steps when the site opens.



Type the reading or upload a meter photo. Readings that go down are rejected, because a meter only counts up.

6400 → 6560

Lakshmi's meter readings

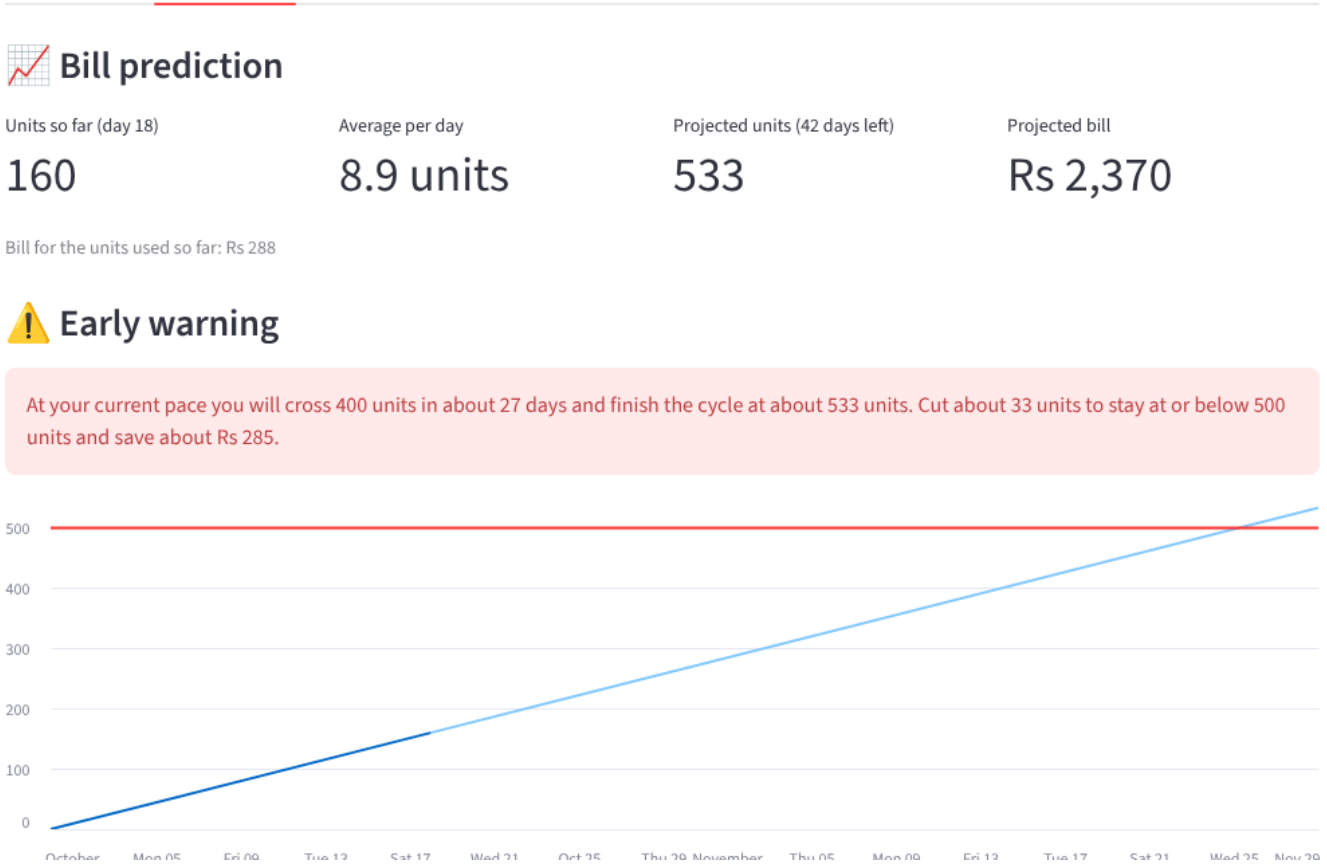
160 units

used in 18 days

8.9 / day

average usage

Prediction and early warning



533 units

predicted for the 60-day cycle

Rs 2,370

predicted bill

~27 days

until crossing 400 units

Predicted units = units so far + units per day × days left

Improvement: a realistic warning

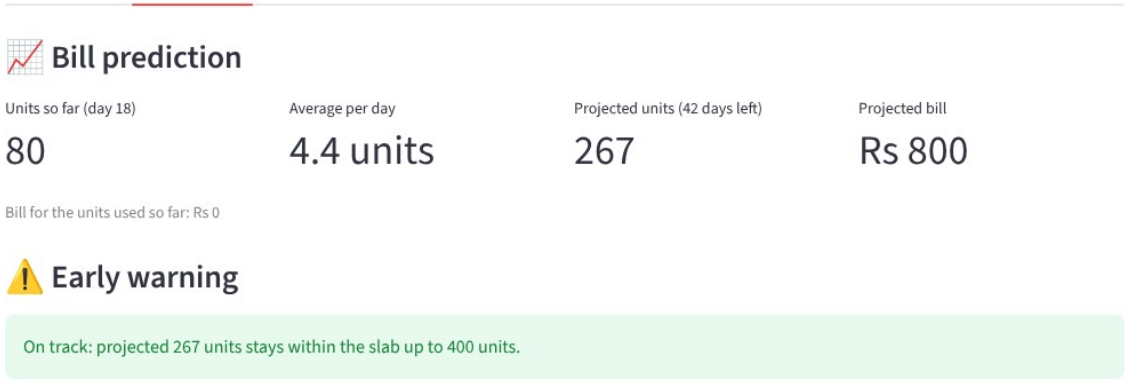
Before

Every home crosses the free 100-unit limit, so the app told almost everyone to cut back to 100 units.

After

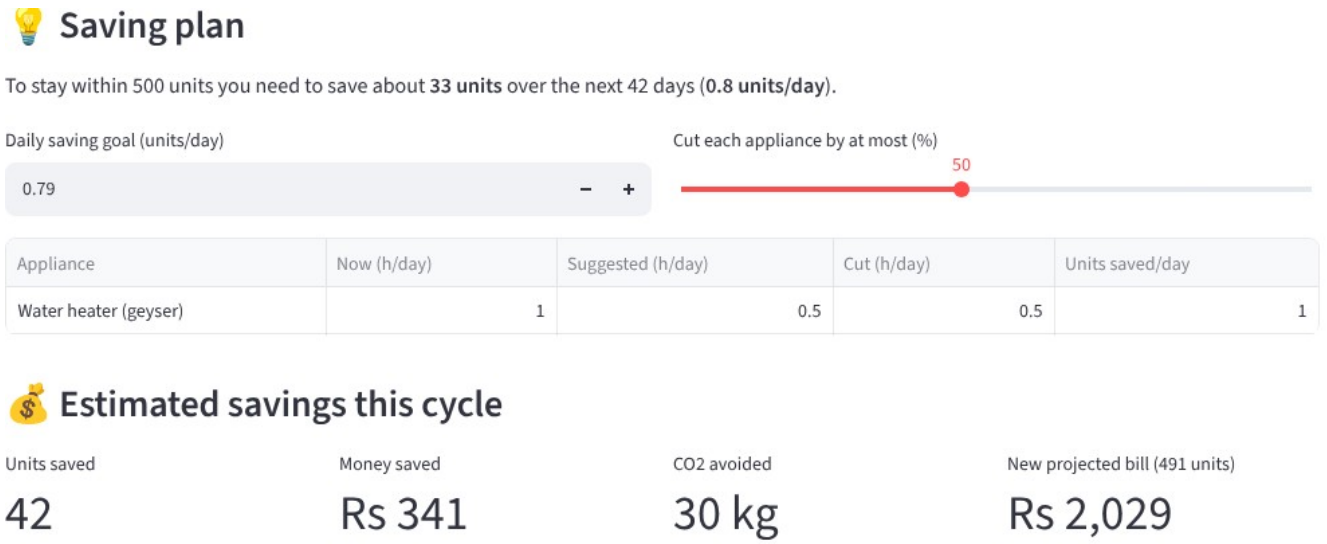
The warning ignores the free-units limit and only checks slabs that cost money (rate > 0).

```
paid_limits = [s.upto for s in tariff.slabs if s.rate
> 0 ...]
```



Sample household Arun (80 units in 18 days) is now correctly shown as on track.

Saving plan: cut the right appliance



- Units per day = watts × quantity × hours ÷ 1000
- Cut the most powerful appliances first
- At most 50% of an appliance's hours; never the fridge

42 units

saved this cycle

Rs 341

money saved

30 kg

CO2 avoided

Water heater 1 h → 0.5 h a day keeps Lakshmi below 500 units.

Results, limitations and future work

Results

- Working web app with database
- Bill matches the slab tariff
- Early warning before costlier slabs
- Saving plan with units, money and CO2
- 33 automated tests pass

Limitations

- Tested with sample household data
- OCR needs clear meter photos
- Prediction assumes steady daily use
- Tariff must be updated when revised

Future work

- Installable mobile app
- User logins per household
- SMS / WhatsApp reminders
- Testing with real homes over a full cycle

Thank you

Questions?

Code: github.com/ZumairaSultana/WATTWATCH